

Exercise 4

$$\begin{aligned}
 (b) \quad \langle v \rangle &= \int_0^\infty v f(v) dv \\
 &= \int_0^\infty v \cdot 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv \\
 &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 \exp\left(-\frac{mv^2}{2k_B T}\right) dv \\
 &\quad \left[\begin{aligned} a &= \frac{m}{2k_B T} \quad \text{so} \quad \int_0^\infty v^3 \exp(-a v^2) dv \\ \text{substitution: } a v^2 &= u \quad \rightarrow \quad \int_0^\infty \exp(-u) \sqrt{\frac{u}{2a}} \frac{du}{2a} \\ \frac{du}{dv} &= 2a v \end{aligned} \right] \\
 &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{1}{2a^2} \\
 &= \frac{4\pi}{2} \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(\frac{2k_B T}{m} \right)^2 \\
 &= \frac{4\pi}{2} \left(\frac{m}{2\pi k_B T} \right)^{3/2-2} \\
 &= 2\pi \sqrt{\frac{2k_B T}{m}} \\
 &= \sqrt{\frac{8k_B T}{\pi m}}
 \end{aligned}$$

c)

$$\lambda = \frac{h}{p} = \frac{h}{m \cdot \langle v \rangle} = \frac{h}{m \sqrt{\frac{8k_B T}{\pi m}}} = \frac{h \sqrt{\pi}}{\sqrt{8k_B T m}}$$

$$\frac{\lambda}{\lambda_T} = \frac{h \sqrt{\pi}}{\sqrt{8\pi k_B T m}} \cdot \frac{\sqrt{2\pi k_B T}}{h} = \frac{1}{2} \Rightarrow \lambda = \frac{\pi}{2} \lambda_T$$

$$\text{Si-O}_2: m = \frac{\mu}{M_A} = \frac{60,0843 \cdot 10^{-3} \text{ mol}}{M_A} \approx 9,98 \cdot 10^{-26} \text{ kg}$$

$$\lambda \rightarrow \lambda_T \approx 0,053 \text{ \AA}$$

$$\text{So we have: } \frac{\lambda_T}{\lambda_{Si-O}} = 0,037 \approx 3,7\%$$

necessary.

Since the thermal de Broglie wavelength is much smaller than the Si-O bond length a quantum mechanical treatment is not